Constellation

Dresden Nuclear Power Station
6500 North Dresden Road
Morris, IL 60450

December 26, 2025

10 CFR 50.73

SVPLTR 25-0071

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Dresden Nuclear Power Station, Units 2 and 3
Renewed Facility Operating License Nos. DPR-19 and DPR-25
NRC Docket Nos. 50-237 and 50-249

Subject: Licensee Event Report 237/2025-002-00, Valid Reactor Protection System Actuation While Shutdown

Enclosed is Licensee Event Report 237/2025-002-00, Valid Reactor Protection System Actuation While Shutdown. This report describes an event being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) for a valid Reactor Protection System actuation.

There are no regulatory commitments contained in this submittal.

Should you have any questions concerning this letter, please contact Ms. Tara Noah, acting Regulatory Assurance Manager, at (779) 231-7447.

Respectfully,

Joseph S. Reda FOR

Hardik Patel
Site Vice President
Dresden Nuclear Power Station

Enclosure: Licensee Event Report 237/2025-002-00

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Dresden Nuclear Power Station, Unit 2	☒ 050 ☐ 052	00237	1 OF 3

4. Title
Valid Reactor Protection System Actuation While Shutdown

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
10	27	2025	2025	- 002 -	00	12	26	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
3	0

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Tara Noah, Acting Manager Site Regulatory Assurance	(779) 237-7447

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☐ No	☒ Yes (If yes, complete 15. Expected Submission Date)		02	15	2026

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 0117 CDT on 10/27/2025, with Unit 2 in Mode 3 at 0 percent power, an actuation of the Reactor Protection System (RPS) occurred during restoration of the Scram Discharge Volume (SDV) high level bypass switch to normal. The cause of the RPS system actuation was a valid high level in the SDV. All control rods had been previously inserted and the RPS system automatically initiated a scram signal as designed when the SDV high level signal was received. The scram was reset in accordance with station procedures at 0121 CDT. An evaluation of this event is currently in progress to determine any contributing causes to the event as well as additional corrective actions.

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) for a valid RPS system actuation.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Dresden Nuclear Power Station	■ 050 ☐ 052	00237	2025	- 002	- 00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION – UNIT 2

General Electric – Boiling Water Reactor, 2957 megawatts thermal rated core power

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

A. CONDITIONS PRIOR TO EVENT

Unit: 2 Event Date: October 27, 2025 Event Time: 0117 CDT

Reactor Mode: 3 Mode Name: Hot Shutdown Power Level: 0%

No systems, structures, or components that were inoperable at the start of the event contributed to the event.

B. DESCRIPTION OF EVENT

At 0117 CDT on 10/27/2025, with Unit 2 in Mode 3 at 0 percent power, an actuation of the Reactor Protection System (RPS) [JC] occurred during restoration of the Scram Discharge Volume (SDV) high level bypass switch to bypass switch to normal. The cause of the RPS [JC] system actuation was a valid high level in the SDV. All control rods had been previously inserted and the RPS [JC] system automatically initiated a scram signal as designed when the SDV high level signal was received. The scram was reset 0121 CDT in accordance with station procedures. ENS report #58008 was transmitted to the NRC at 0750 CDT on 10/27/2025. An evaluation of this event is currently in progress.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A).

C. CAUSE OF EVENT

The cause of the event was a valid high level in the SDV. An evaluation of the event is currently in progress. A supplemental report will be submitted with the conclusions of this evaluation.

D. SAFETY ANALYSIS

The RPS [JC] monitors reactor [RCT] core parameters. If these parameters exceed predetermined limits, the RPS [JC] logic will initiate a scram. The SDV is used to limit the loss of, and contain, the reactor vessel [RPV] water from all Control Rod Drives (CRDs) [AA] during a scram. During normal operation, the SDV is empty with open drain [DRN] and vent valves [VTV]. On a scram signal, the RPS [JC] is deenergized and the SDV pilot valves [V] vent the discharge volume valve [V] operators, causing them to close. All systems and components responded as expected to the valid high SDV level signal. Additionally, all CRDs [AA] had been previously inserted. There was no impact to the health and safety of the public from this event.

NARRATIVE

E. CORRECTIVE ACTIONS

At 0121 CDT on 10/27/2025, the scram was reset in accordance with station procedures. Additional corrective actions will be determined by the evaluation currently in progress.

F. PREVIOUS OCCURENCES

A 5-year internal corrective action program and Licensee Event Report (LER) database review was performed for Dresden Nuclear Power Station (DNPS). No previous occurrences of valid RPS [JC] actuation were found. Other events identified during the evaluation in progress will be discussed in the supplemental report.

G. COMPONENT FAILURE DATA

Not applicable